Collapse Pathways

A comparative historical collapse-pattern analysis and global fragility framework

Professional presentation dossier

Prepared for portfolio / presentation use
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Executive Summary

Collapse Pathways is an uncertainty-aware comparative research project that turns diverse historical interpretations into a structured dataset spanning 15 major cases of state, imperial, and regional breakdown. Rather than treating collapse as a single-cause event, the project identifies recurrent clusters of political, economic, social, environmental, military, and institutional stress. The final stage extends those historical findings into a present-day Global Fragility Index with scenario analysis.

- Dataset scope: 174 coded time-window rows covering 15 historical cases from the Western Roman Empire to Yugoslavia.
- Strongest historical signals: territorial loss, political fragmentation, legitimacy crisis, fiscal crisis, urban decline, and internal conflict.
- Uncertainty is explicit: difficult-to-compare variables are retained but marked as unknown (coded 9) rather than forced into false precision.
- Present-day extension: baseline Global Fragility Index of 73.35 / 100, with conservative and high-stress scenarios at 55.25 and 91.45.

1. Introduction

Historical collapse is often studied case by case, while public discussions of present-day instability often leap too quickly into prediction. This project is designed to sit between those extremes. It asks how collapse patterns can be studied comparatively, in a disciplined way, without pretending that ancient societies and the contemporary world are identical.

The project therefore uses a transparent coding framework and emphasizes three principles throughout: comparability, uncertainty handling, and interpretability. The result is not an “apocalypse model,” but a structured way to examine how breakdown pathways recur and how they might overlap with contemporary global conditions.

2. Research Question and Objectives

- Which factors recur most consistently across major historical collapses and severe systemic transformations?
- Which combinations of factors differentiate lower-collapse from higher-collapse rows?
- Which factors are strongly supported across the dataset, and which remain evidence-limited?
- How strongly do historically recurrent patterns overlap with selected present-day global stress conditions?

3. Project Design

The project follows a staged workflow from historical coding to present-day scenario analysis:

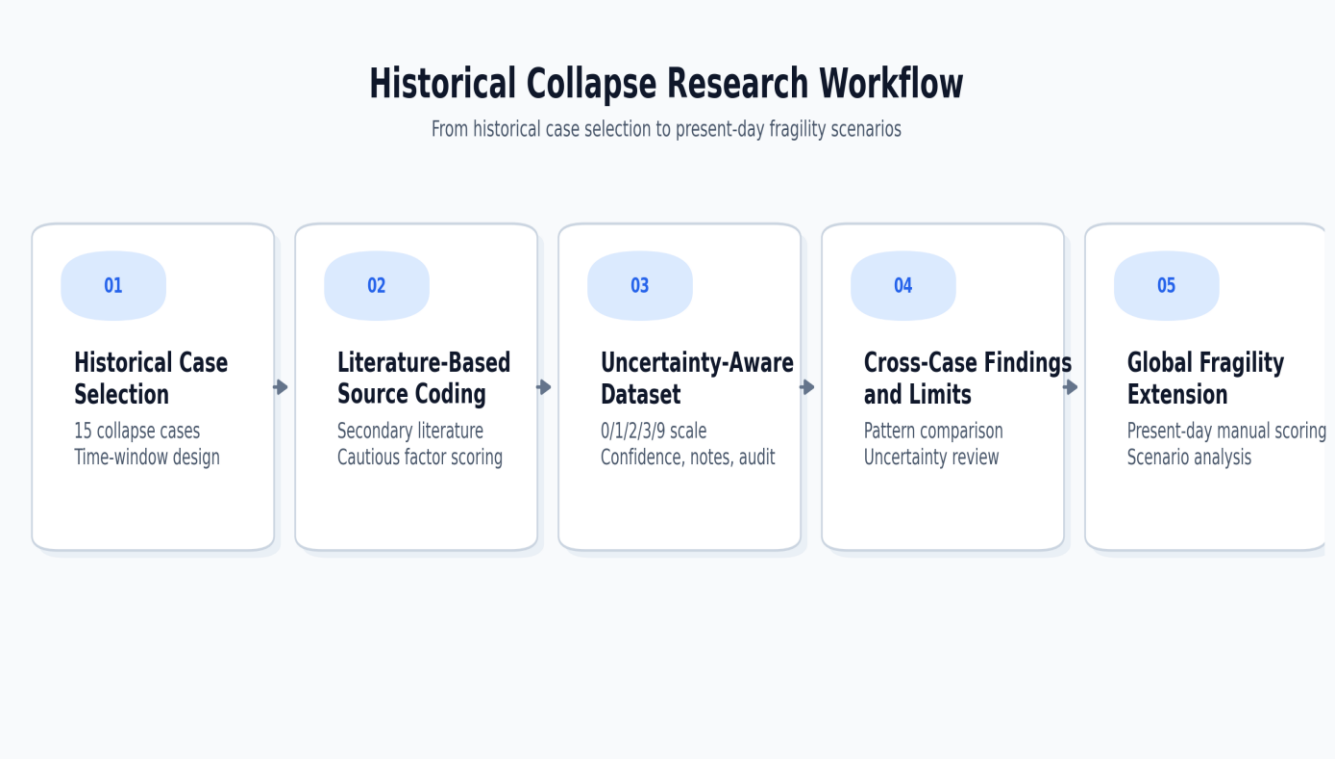

Figure 1. High-level workflow from historical case selection to present-day global fragility scoring.

Historical cases included

Western Roman Empire; Bronze Age Collapse States; Maya; Akkadian Empire; Khmer Empire; Han Dynasty Crisis; Byzantine Decline; Sassanian Empire; Easter Island; Classic Mesopotamian States; Ming/Qing Transition; Aztec Empire; Inca Empire; Soviet Union; Yugoslavia.

Factor framework

Category	Variables
Political	political_fragmentation, elite_conflict, succession_crisis, legitimacy_crisis, administrative_overload, corruption_governance_failure
Social	social_inequality, social_unrest_rebellion, demographic_pressure, migration_pressure, ethnic_sectarian_fragmentation, urban_decline
Economic	fiscal_crisis, taxation_extraction_pressure, trade_disruption, inflation_currency_instability, resource_dependency, agricultural_decline
Environmental	drought_climate_stress, flood_environmental_shock, temperature_anomaly, ecological_degradation, food_insecurity
Military / geopolitical	external_invasion_pressure, civil_war_internal_conflict, military_overstretch, territorial_loss
Institutional / resilience	institutional_rigidity, adaptive_capacity, logistics_food_storage_resilience, alliance_network_strength, recovery_capacity

Scoring scale: 0 = absent / no evidence, 1 = low, 2 = moderate, 3 = high, 9 = unknown / insufficient evidence / debated interpretation.

4. Data Construction and Uncertainty Handling

Each row in the dataset represents one case in one time window. The coding process translates literature-based expert interpretations into structured factor scores. Because not all cases support the same level of precision, uncertainty is treated as an explicit feature of the dataset rather than a nuisance to hide.

- Rows also track phase_label, collapse_outcome, collapse_within_next_window, data_confidence, source_count, and notes.
- Unknowns are coded as 9, preserving interpretive caution for debated or weakly evidenced variables.
- A sanity-review pass was used to catch timeline inconsistencies, repeated-note blocks, and undercoded or overconfident rows.
- The final dataset is intended for exploratory pattern analysis, not final historian-grade truth claims.

5. Main Historical Findings

Cross-case comparison shows that historical collapse is best understood as a multi-factor breakdown syndrome rather than a monocausal event.

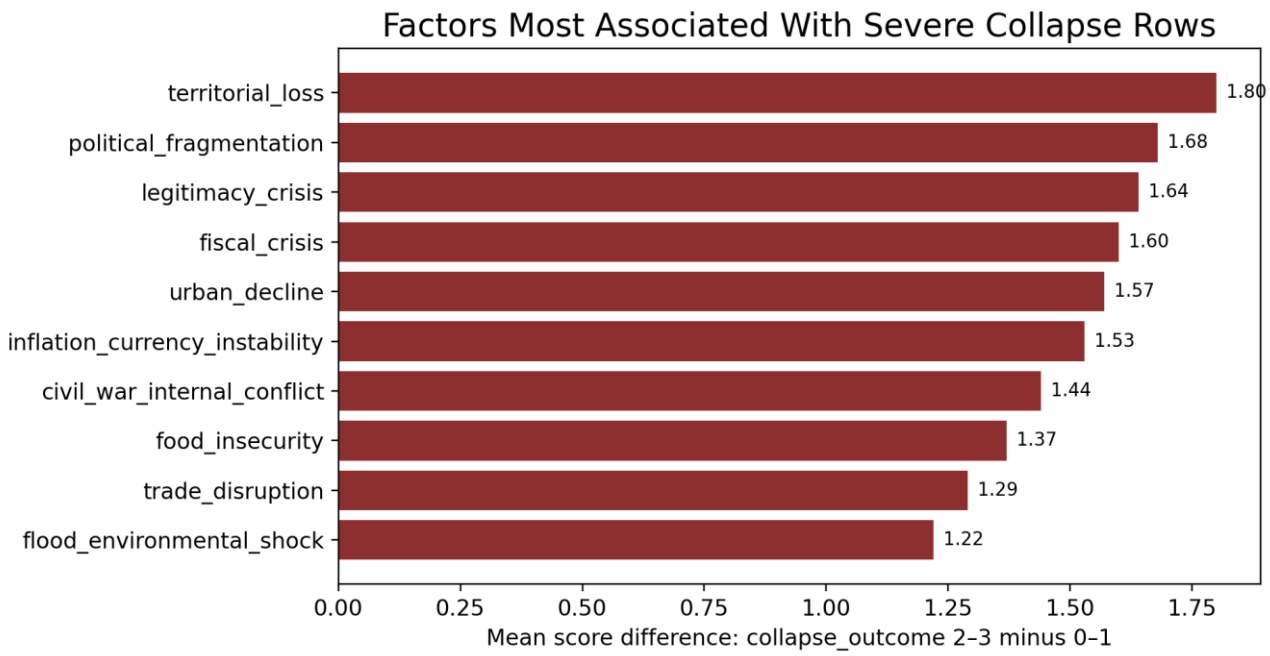

Figure 2. Factors that rise most sharply when moving from lower-collapse rows to severe-collapse rows.

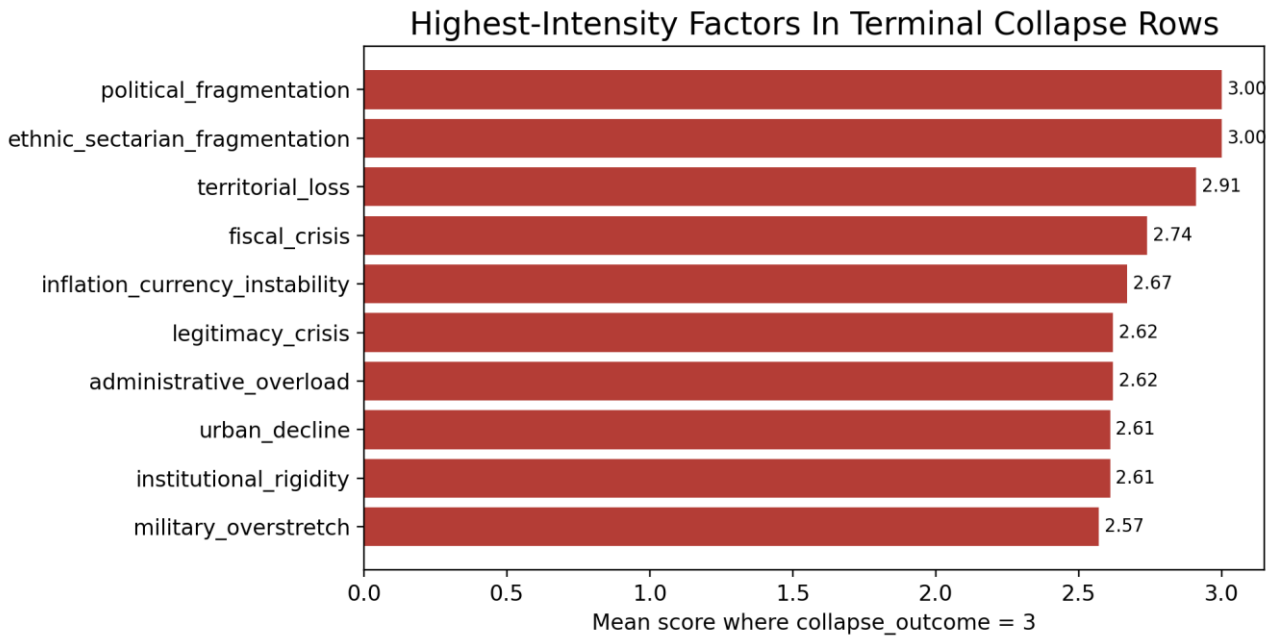

Figure 3. Highest-intensity factors within rows coded as terminal collapse or fragmentation.

Interpretation

- The strongest differentiators between lower-collapse and higher-collapse rows are territorial loss, political fragmentation, legitimacy crisis, fiscal crisis, urban decline, and civil/internal conflict.
- Terminal-collapse rows are especially intense on fragmentation, territorial erosion, fiscal stress, legitimacy strain, administrative overload, and institutional rigidity.
- The pattern supports a clustered explanation of collapse: political, territorial, and fiscal breakdowns tend to reinforce one another.

6. Evidence Limits and Interpretive Caution

Not all variables are equally comparable across cases. Some are important in particular contexts but remain difficult to code consistently across all 15 cases.

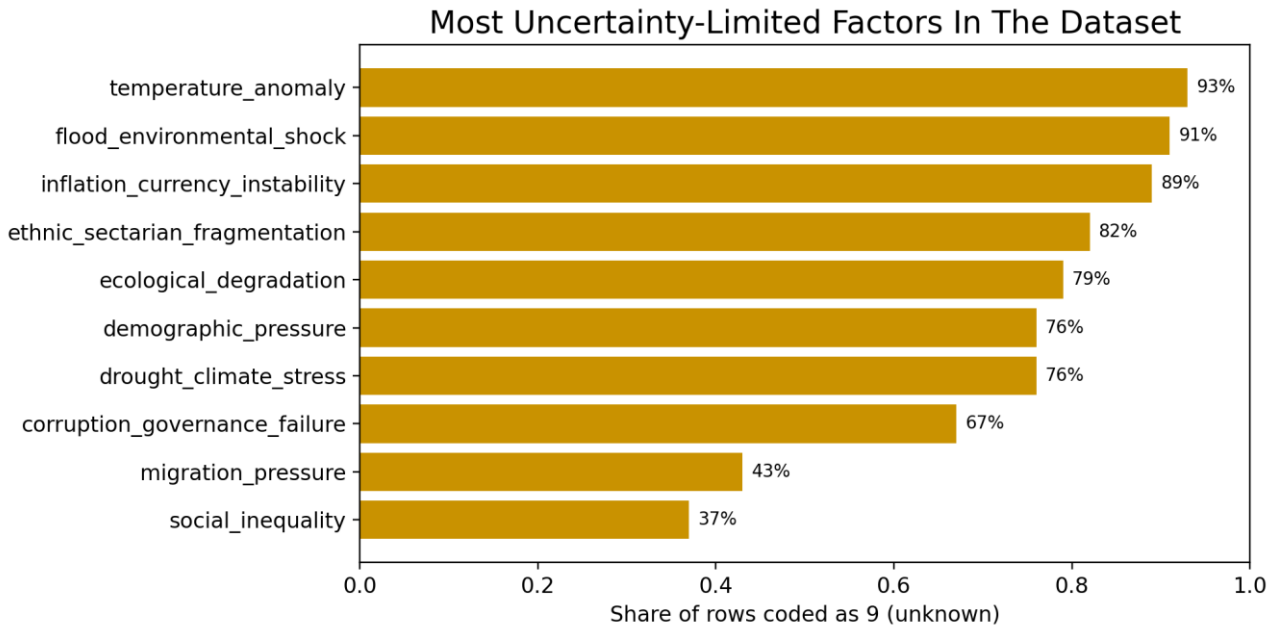

Figure 4. Variables most limited by uncertainty, shown as the share of rows coded 9 = unknown.

- Environmental and demographic variables matter in important cases, but they remain among the most uncertainty-limited parts of the dataset.
- Cases such as Bronze Age Collapse States, Easter Island, Maya, Classic Mesopotamian States, and Inca Empire remain more heterogeneous than modern state-collapse cases.
- The strongest claims should therefore rely most heavily on factors that remain visible despite uncertainty.

7. Present-Day Global Extension

The project includes a manual present-day extension that converts historical findings into a Global Historical Pattern Overlap / Fragility Index. This extension does not predict collapse. Instead, it measures how strongly current global conditions overlap with the types of stress patterns that recur in the historical dataset.

Global scoring framework

- Core present-day factors include legitimacy crisis, geopolitical instability, adaptive capacity/resilience, political fragmentation / polarization, trade disruption, conflict pressure, fiscal stress, urban social stress, institutional rigidity, food insecurity, resource dependency, and environmental stress.
- Historical weights are held fixed; present-day scores are entered manually and interpreted through conservative, baseline, and high-stress scenarios.
- Resilience factors offset, rather than intensify, the net fragility score.

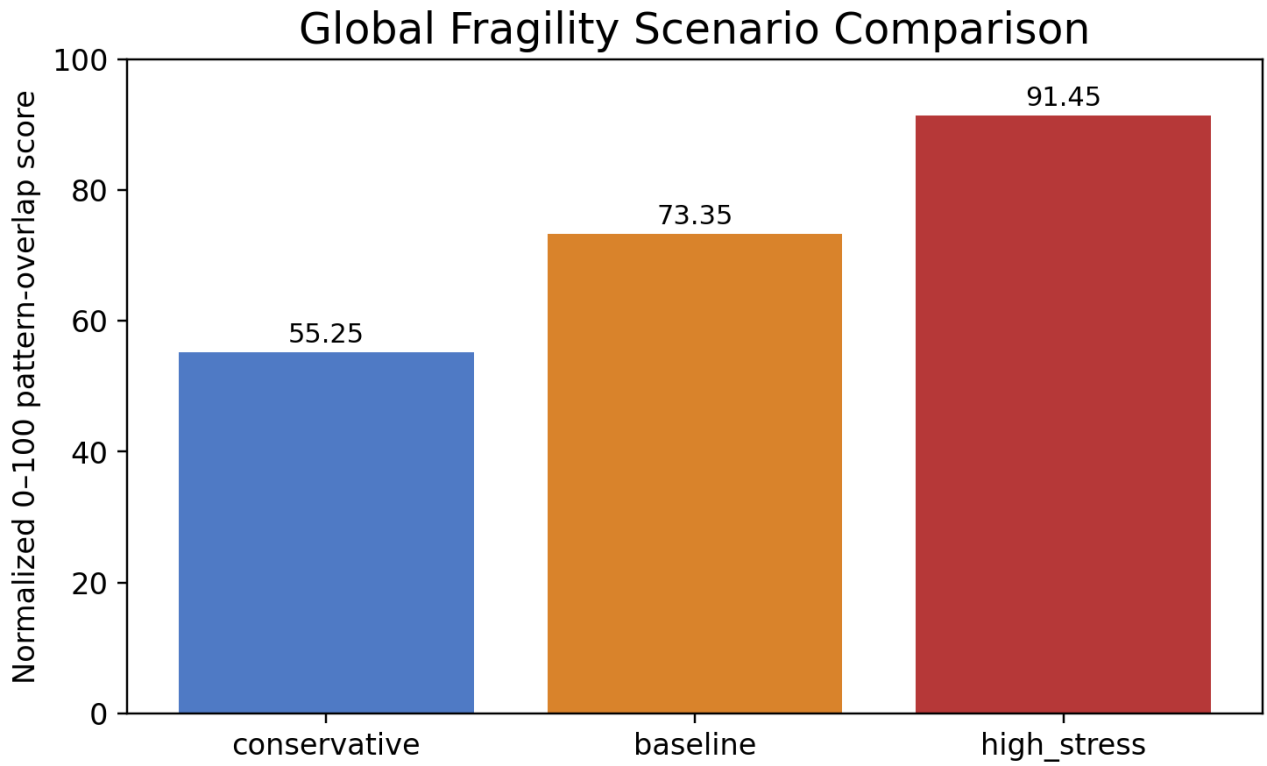

Figure 5. Scenario comparison for the present-day global fragility extension using fixed historical weights and varying current-world scoring assumptions.

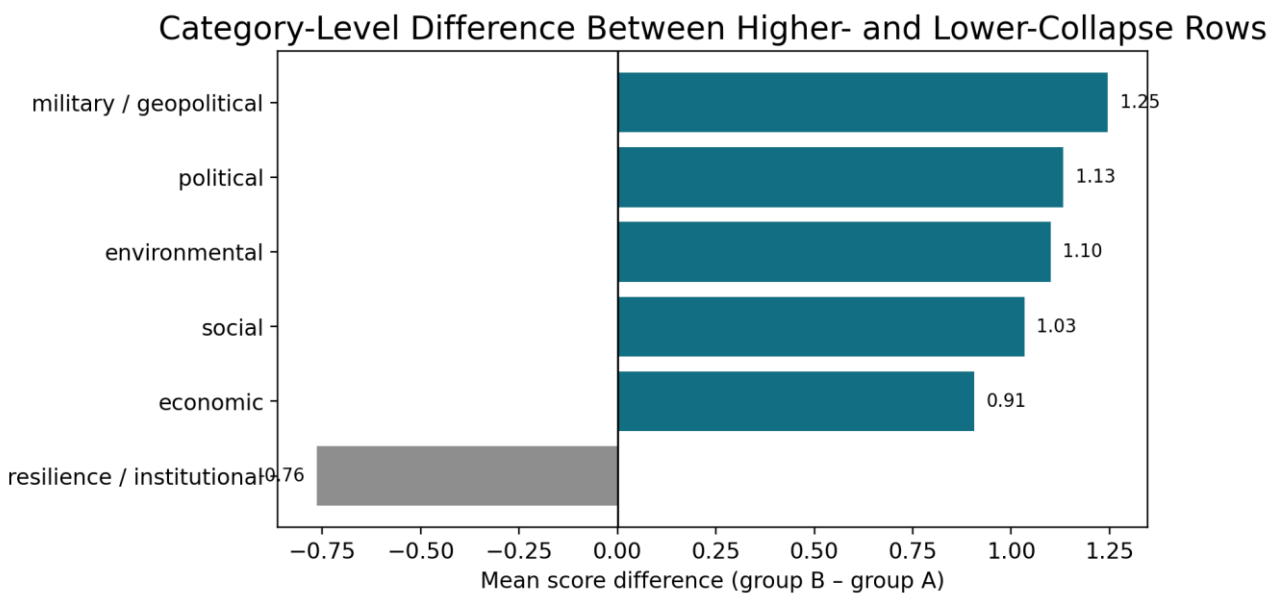

Figure 6. Category-level difference between higher- and lower-collapse rows; positive values rise with more severe collapse, while resilience/institutional factors offset fragility in the global extension.

8. Scenario Results

Scenario	Score	Interpretation
Conservative	55.25	Moderate fragility

Baseline	73.35	High fragility
High-stress	91.45	Severe systemic stress

The scenario spread demonstrates that the framework is best used for structured scenario analysis rather than a single precise “true” number. The baseline score places the current world in a high-fragility overlap range, while the conservative and high-stress cases show how sensitive the output remains to present-day scoring assumptions.

9. What the Project Can and Cannot Support

What it can support

- A transparent comparative account of which historical stress patterns recur across major breakdowns.
- A cautious, uncertainty-aware summary of which variables are most robust versus evidence-limited.
- A scenario-based present-day overlap framework that helps structure discussion of global fragility.

What it cannot support

- A literal prediction of collapse or a timetable for future breakdown.
- A claim that ancient and modern societies are directly interchangeable.
- A final authoritative historical truth table for every case and variable.

10. Bibliography

Selected historical sources used to construct the coding framework and case notes. Full source inventory is available in the repository.

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11. Closing Note

Collapse Pathways demonstrates a practical way to turn qualitative historical interpretation into a disciplined analytical workflow. Its main contribution is not prediction, but structured comparison: it shows which stress clusters recur historically, where uncertainty remains high, and how those patterns can be translated into a present-day fragility framework without overstating certainty.